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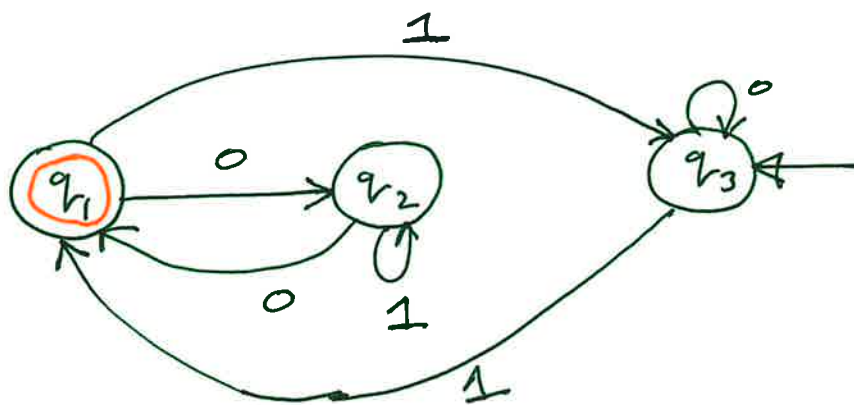
CS340

Lecture-9

NFA to R.E.
(Examples)

Example:

Let M be an automaton as follows, which recognizes 0-1 strings ~~s.t.~~ x s.t. in binary rep ~~of~~ x denotes a number $1 \bmod 3$.



i, j	$k=0$	$k=1$	$k=1$ simplified	$k=2$
1 1	ϵ	$\epsilon + \epsilon$	ϵ	$\epsilon + 0(1+00)^*0$
1 2	0	$0 + 0$	0	$0 + 0(1+00)^*(1+00+\epsilon)$
1 3	1	$1 + 1$	1	$1 + 0(1+00)^*01$
2 1	0	$0 + 0.\epsilon$	0 + 0	$0 + (1+00)^*0$
2 2	$1 + \epsilon$	$1 + \epsilon + 0.0$	$1 + 00 + \epsilon$	$\epsilon + 00 + 1 + (1+00)^*(1+00+\epsilon)$
2 3	ϕ	0.1	01	$01 + (1+00)^*.1$
3 1	1	$1 + 1.\epsilon$	1	$1 + 10(1+00)^*0$
3 2	ϕ	1.0	10	$10 + 10(1+00)^*(1+00+\epsilon)$
3 3	$0 + \epsilon$	$0 + \epsilon + 1.1$	$0 + 11 + \epsilon$	$0 + 11 + \epsilon + 10(1+00)^*01$

Entries in the table are R_{ij}^k

$$(R_{11}^0)^* = \epsilon^* = \epsilon$$

$$(R_{22}^1)^* = (1 + 00 + \epsilon)^* = (1+00)^*$$

R_{ij}^k table continued

i, j	$k=2$ simplified
1 1	$E + 0(1+00)^*0$
1 2	$0(1+00)^*$
1 3	$1 + 0(1+00)^*01$
2 1	$(1+00)^*.0$
2 2	$(1+00)^*$
2 3	$(1+00)^*01$
3 1	$1 + 10(1+00)^*0$
3 2	$10(1+00)^*$
3 3	$E + 0 + 11 + 10(1+00)^*01$

We are interested in R_{31}^3

$$R_{31}^3 = R_{31}^2 + R_{33}^2 (R_{33}^2)^* R_{31}^2$$

$$= [E + (R_{33}^2)^+] R_{31}^2 = (R_{33}^2)^* R_{31}^2$$

$$= (E + 0 + 11 + 10(1+00)^*01)^* (1 + 10(1+00)^*0)$$

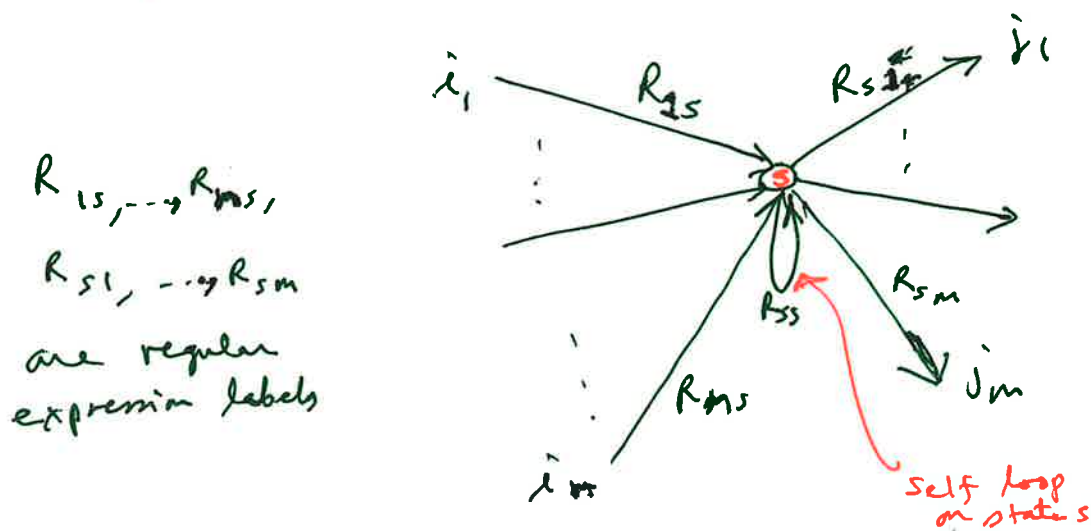
$$= (0 + 11 + 10(1+00)^*01)^* (1 + 10(1+00)^*0)$$

Another algorithm (to convert NFA to a RE)

This algorithm works by successively eliminating states from the automaton. The elimination should satisfy the condition, that between any two states in the reduced ~~graph~~ transition diagram the set of words on paths from one state to another are the same before and after the state elimination.

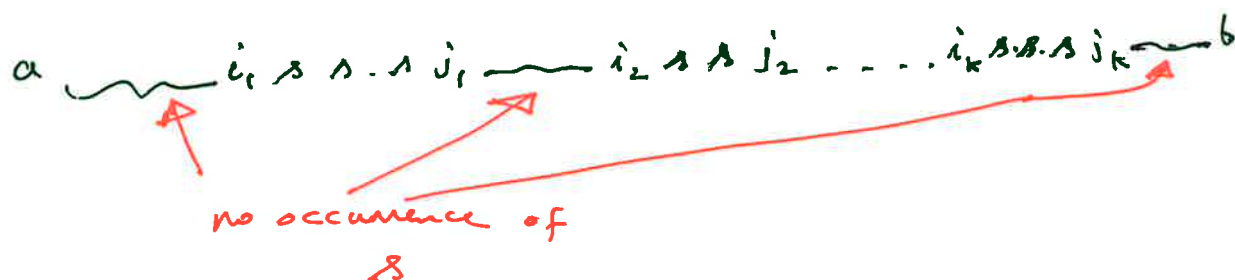
The reduced transition diagram has edges labeled by regular expressions, rather than just by a letter of Σ .

Let us consider eliminating state s , which has self loop labeled R_{ss} . ~~and any~~ Consider all incoming and edges and all outgoing edges for s .



After eliminating s , for $u \in \{1, \dots, n\}$ and $v \in \{1, \dots, m\}$ a new edge from i_u to j_v labeled $R_{us}(R_{ss})^* R_{sv}$ is created.

To see preservation of the language, let a, b be vertices in the transition diagram after eliminating state s . Consider any path between a and b before eliminating s , it in general looks like



Also note that $i_1, j_1 \neq s \dots i_k, j_k \neq s$

~~(note that $j_u = i_{u+1}$ etc. is possible)~~

Note that $\underbrace{i_u \rightsquigarrow \dots \rightsquigarrow j_u}_{\text{words in}} \dots$ are contained in expression $R_{us}(R_{ss})^* R_{su}$

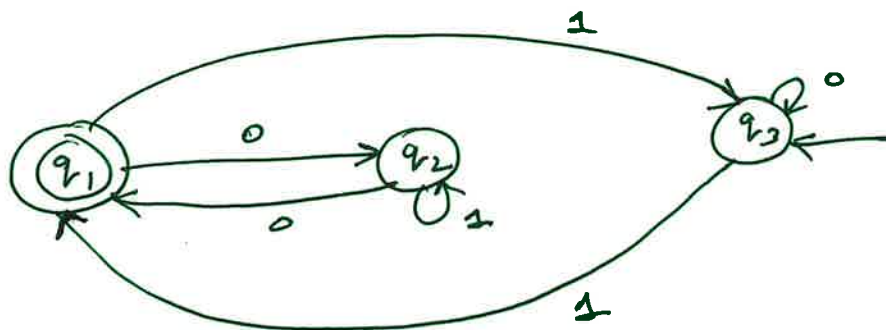
so path segment $\underbrace{i_u \rightsquigarrow \dots \rightsquigarrow j_u}$ can be replaced by

new edge $i_u \xrightarrow{R_{us}(R_{ss})^* R_{su}} j_u$

similarly for other segments involving s .

Example

We now run the state elimination algorithm on the previous example.



We are interested in paths from q_3 to q_1 . Therefore state q_2 may be eliminated.

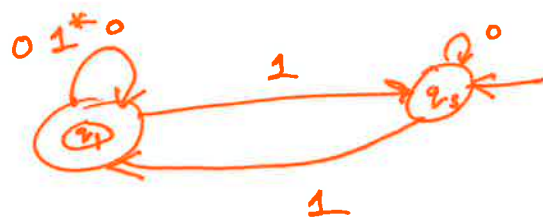


q_2 has exactly one incoming and exactly one outgoing edge (apart from the self loop)

Eliminating q_2 introduces new edge labeled

$0 1^* 0$ from q_1 to q_1

The new state diagram is



All paths from q_3 to q_1 are given by

$$(0 + 1(0 1^* 0)^* 1)^* 1 (0 1^* 0)^*$$

This is the RE we were looking for.

One may note that RE obtained by two different methods look quite different even if there are supposed to define the same language.

Can we show equivalence of

$$(0 + 11 + 10(1+00)^*01)^*(1 + 10(1+00)^*0)$$

and $(0 + 1(01^*0)^*1)^*1(01^*0)^*$?

A first step is to relate $(1+00)^*$ and $(01^*0)^*$ in some way. Expanding $(01^*0)^*$, we see that it has words like $01^*001^*001^*\dots001^*0$. This suggests the following.

Exercise show that $(01^*0)^* = 0(1+00)^*0 + \epsilon$

Plugging in the result of above exercise in the second expression above (the one obtained by state elimination) we get

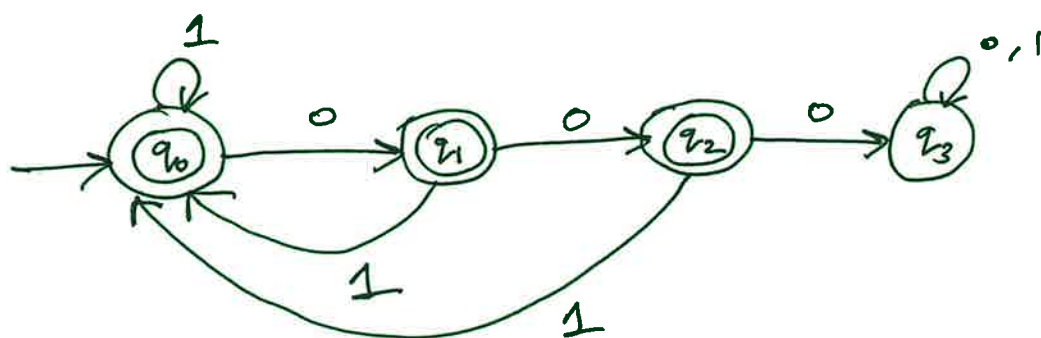
$$(0 + 1[0(1+00)^*0 + \epsilon]1)^*1(0(1+00)^*0 + \epsilon)$$

$$= (0 + 10(1+00)^*01 + 11)^*1(10(1+00)^*0 + 1)$$

which is the same as the first expression.

Example

As another example consider the following DFA.

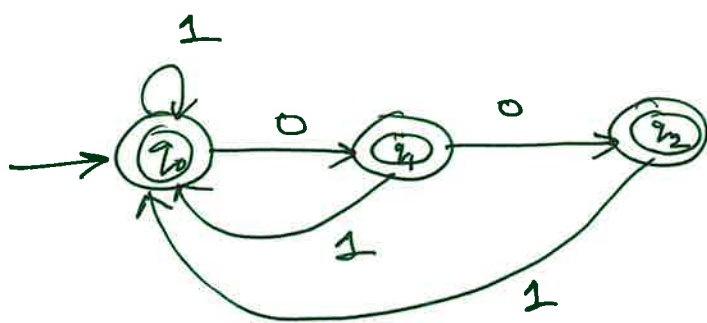


This recognizes the language of binary words which do not have 3 consecutive 0's.

q_3 is a trap state, it does not serve a useful purpose towards recognition of this language but needs to be there because a DFA requires its transition to be total (defined on all elements of $Q \times \Sigma$).

We can avoid this state if we are required to construct only a NFA.

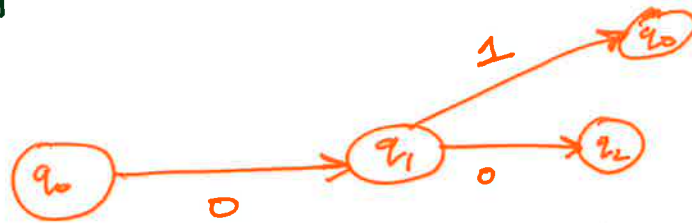
In other words, NFA below recognizes the same language.



We convert this to equivalent RE. we are interested in the three sets of paths: q_0 to q_0 , q_0 to q_1 and q_0 to q_2 . We need to deal with each separately.

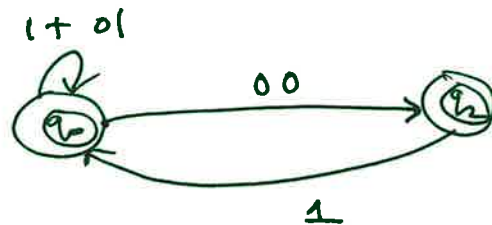
We first construct RE for ~~the~~ language of words from q_0 to q_0 .

Eliminating q_1



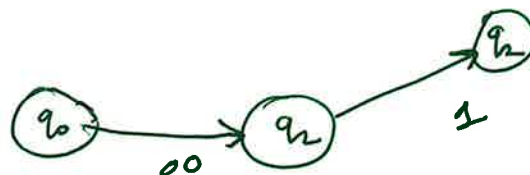
This introduces two new edges. q_0 to q_0 labeled 01 and q_0 to q_2 labeled 00 .

Reduced automaton is



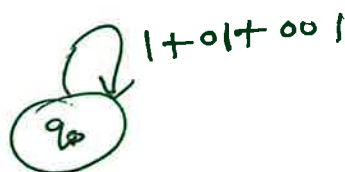
~~Set of~~

Eliminating q_2



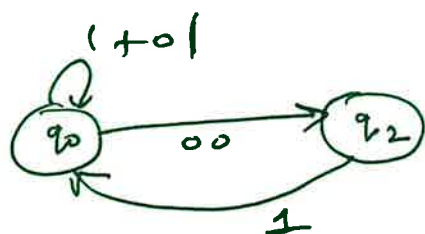
This introduces a new edge q_0 to q_2 labeled 001

Resulting automaton is



language of words
The ~~set~~ of all paths from q_0 to q_0 is therefore $(1 + 01 + 001)^*$

For paths from q_0 to q_2 , we can use the reduced automaton ~~in the~~ above, obtained by eliminating q_1 .



from this we get the set of all paths from q_0 to q_2 as

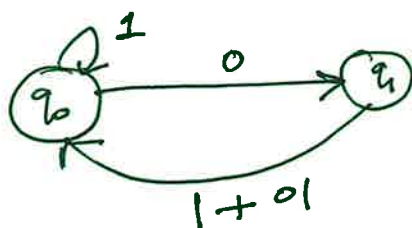
$$(1+01+001)^*00$$

For q_0 to q_1 language, we eliminate q_2 from the original ~~reduced~~ automaton.



This introduces a new edge labeled 01 from q_1 to q_0 .

The resulting automaton is



language of words from q_0 to q_1 is

$$\begin{aligned} & (1+0(1+01))^*0 \\ &= (1+01+001)^*0 \end{aligned}$$

Combining all three set of paths we get RE

$$(1+01+001)^* + (1+01+001)^*00 + (1+01+001)^*0$$

$$= (1+01+001)^*(\epsilon + 00 + 0) \quad [\text{By distributive law}]$$

This is the same expression we saw in the last class.